# Atrey Desai

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Google Scholar

## Education

**University of Maryland, College Park** *Exp. Graduation: May 2027*
*B.S. in Computer Science (Honors) & B.A. in Linguistics (Minor in Korean Studies)* * – in progress
Advisors: Prof. Rachel Rudinger & Prof. Jordan Boyd-Graber
*Coursework:* Natural Language Processing (Grad), Commonsense Reasoning (Grad), Mechanistic Interpretability (Grad)*, Machine Learning, Data Science, Parallel Computing, Computer Systems, Syntax, Semantics*, Phonetics, Psycholinguistics

## Experience

**University of Maryland, College Park** *May 2024 – present*
*Undergrad Researcher, CLIP Lab*
Advisors: Prof. Rachel Rudinger & Prof. Jordan Boyd-Graber

- Quantified differential LLM susceptibility to human-written vs. LLM-generated MCQs to assess if questions contain unintended artifacts and are solvable without full context; created tools to isolate distractor-choice provenance effects and improve the reliability of human synthetic data for model evaluation and distillation.
- Constructed an adversarial benchmark to test VLM capabilities in detecting out-of-context (OOC) video-based misinformation on social media based on multimodal clues and user interactions to probe reasoning under deceptive framing.
- Applied causal interpretability techniques to demonstrate that LMs require significantly more data than humans to acquire general syntactic representations, revealing heightened sensitivity to item-level and construction-level variation.

**The University of Texas at Arlington** *Feb 2024 – present*
*Visiting Researcher, ACL2 Lab & NSF*
Advisor: Prof. Kenny Zhu

- Designed AniVoice-cat, a dataset of 26,000+ annotated cat vocalizations from 250+ hours of video and identified 57 unique cat phones, establishing a foundational resource for computational research in non-human communication.
- Improved transcription pipeline using PANNs and HuBERT models to 96% accuracy with 93% top-5 accuracy in action recognition, setting a new state-of-the-art for automated animal vocalization analysis.

**Learn Prompting** *May 2025 – Jan 2026*
*Member of Technical Staff (San Francisco, CA)*

- Ran Hackaprompt, the world's largest red-teaming hackathon, overseeing user engagement and technical challenges.
- Researched the robustness of AI safety judges against CBRNE-related adversarial content.

**University of Maryland, College Park** *Dec 2023 – Aug 2024*
*Researcher, FIRE Sustainability Analytics Lab*
Advisor: Prof. Thanicha Ruangmas

- Developed Python pipeline for environmental impact assessments of U.S. emissions, enabling more efficient policy.
- Drafted a framework to guide evidence-based policymaking on climate restoration strategies.

**Brown University** *Dec 2020 – June 2023*
*Researcher, Reinforcement Learning at Brown Group*
Advisor: Prof. Michael Littman

- Developed a custom RL environment that empowered non-technical users to programmatically solve complex tasks by defining reward functions and specifying agent behavior, reducing task setup time.
- Published and presented research at two top-tier workshops (AAAI, RLDM), demonstrating how human-readable interfaces enable fine-grained control during inference and improve AI-human interaction in robotics.

## Publications

### Preprints

* – equal contribution

1. A Preview of Computational Animal Linguistics
   Under Review (ACM Computing Surveys)
   Tuan Dang*, **Atrey Desai***, Hridayesh Lekhak*, Tirza Panunto*, Lindsay Pike*, Theron Wang*, and Kenny Zhu*

### Publications

5. Filling in the Mechanisms: How do LMs Learn Filler-Gap Dependencies under Developmental Constraints?
   ACL 2026 (Findings), TLS 2026 (Oral)
   **Atrey Desai** and Sathvik Nair

4. BenchMarker: An Education-Inspired Toolkit for Highlighting Flaws in Multiple-Choice Benchmarks
ACL 2026
Nishant Balepur, Bhavya Rajasekaran, Jane Oh, Michael Xie, **Atrey Desai**, . . . , Jordan Boyd-Graber

3. Test-Time Reasoners Are Strategic Multiple-Choice Test-Takers
ACL 2026 (Oral)
Nishant Balepur, **Atrey Desai** and Rachel Rudinger

2. Language Models Generate Multiple-Choice Questions with Artifacts
MASC-SLL 2025
**Atrey Desai**, Nishant Balepur and Rachel Rudinger

1. Reinforcement Learning As End-User Trigger-Action Programming
IML Workshop @ AAAI 2022; RLDM 2022
Chace Hayhurst, Hyojae Park, **Atrey Desai**, Suheidy De Los Santos and Michael Littman

## Selected Awards and Honors

- **CRA NSF CISE Student Funding Award** ($9,000) *2026*
- **Department of Computer Science Professional Development Fund** ($500) *2026*
- **SPIRE Research Grant** ($3,000) *2025*
- **Omicron Delta Kappa Top 10 Freshman** *2024*
- **CMSC & ARHU Dean's List** *2023 – 2026*
- **UMD President's Scholarship** ($50,000) *2023*
- **NMSC National Merit Scholarship** ($4,000) *2023*
- **Catherine Yang Scholarship** ($1,000) *2023*

## Talks

- **Filler-Gap Dependencies under Developmental Constraints in LMs**
  - Texas Linguistics Society *Feb 2026*
  - UMD Computational Cognitive Science Reading Group *Feb 2026*
- **Adaptor Grammars and Neural Networks for Feline Lexical Discovery**
  - University of Maryland, College Park *Nov 2024*
  - UT Arlington, Department of Computer Science *July 2024*

## Professional Responsibilities

- **Subreviewer**, ACL Rolling Review (ARR) *2025 – present*
- **Member**, University Research Advisory Board *2025 – present*
- **Vice Chair**, Computer Science Department Council *2025 – present*
  *Appointed by faculty and student body; represent 4,200+ CS undergraduates.*
- **Senior Member**, FIRE Student Leadership Council *2024 – present*
  *Represented 1000+ peers, ran events & workshops, and reformed class curricula.*
- **University Ambassador** *2024 – present*
  *Represented the CS Dept. & CMNS College at admissions events and to official guests.*
- **Mentorship**
  - **Office of Undergraduate Research:** Juan Cortés, Kemisola Benson, Vivian Akpala *2025*
  - **Technica Mentoring:** Savya Miriyala, Tanya Grover, Jessica Ononye, Nakshatra Hiray *2024*
  - **MSET Robotics:** Workshop organizer and curriculum designer *2020 – 2022*

## Skills

- **Languages:** Python, Java, JavaScript/HTML/CSS, R, MATLAB.
- **Libraries/Frameworks:** Huggingface (Datasets, Transformers), NLTK, PyTorch, Selenium, BeautifulSoup.
- **Tools:** Git, Docker, GCP, Google Vertex AI, VS Code.
- **Natural Languages:** English (Native), Gujarati (Native), Spanish (Intermediate), Korean (Beginner).